Exercise 01: Team Formation Report

- **Course:** Computational Thinking (CS252 - Class 25A01)
- **Student ID:** 25125048
- **Student Name:** Lý Trần Thái
- **Team Size:** 7 members
- **Team Leader:** Nguyễn Tấn Minh (25125027)
- **Group Repository:** https://github.com/callmelucian/cs252-final-project
- **Submission Date:** October 17, 2026

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1. Team Overview & Official Registration

In accordance with course guidelines for the final course project, our group has formed a collaborative team of 7 members. The team information has been officially registered via the instructor's submission form.

Team General Information

- **Course Code:** CS252 (Computational Thinking)
- **Class:** 25A01
- **Team Size:** 7 members
- **Team Leader:** Nguyễn Tấn Minh
- **Shared Workspace / Repository:** cs252-final-project

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2. Member Roster & Role Assignments

The table below summarizes each member's identity, permanent GitHub account, and their primary role designated within the course project:

Student ID	Full Name	GitHub Username	Project Role	Contact Email
25125027	Nguyễn Tấn Minh	@callmelucian	Team Leader	25125027@student.hcmus.edu.vn
25125084	Trần Gia Huy	@FubuGold	Main System Design / Architect	25125084@student.hcmus.edu.vn
25125050	Bùi Đỗ Thiên Ân	@Gasmask-Chan	Algorithm & Logic Developer	25125050@student.hcmus.edu.vn
25125014	Nguyễn Đức Huy	@SpheraXX	Front-end Developer	25125014@student.hcmus.edu.vn
25125048	Lý Trần Thái	@LyTranThai	Core Developer & Data Integration	25125048@student.hcmus.edu.vn

Student ID	Full Name	GitHub Username	Project Role	Contact Email
25125056	Trần Đăng Khoa	@trankhoa8581	Back-end Developer	25125056@student.hcmus.edu.vn
23125054	Ngô Tấn Đạt	@diamondzone1	Security & Data Engineer	23125054@student.hcmus.edu.vn

3. Member Strengths, Weaknesses & Contribution Commitments

During our team formation discussions, we analyzed the unique profile of each member to balance responsibilities effectively and prevent project bottlenecks:

1. Nguyễn Tấn Minh (25125027) - Team Leader

- **Strengths:** High organizational acumen, decisive task management, proficient with agile sprint workflows, clear communication with faculty and peers.
- **Weaknesses:** High administrative workload occasionally limits availability for deep, uninterrupted feature coding.
- **Agreed Contributions:** Overseeing project milestones, facilitating sprint check-ins, tracking deliverables on shared sheets, acting as primary contact with course instructors, and arbitrating technical decisions.

2. Trần Gia Huy (25125084) - Main System Design / Architect

- **Strengths:** Strong high-level architectural intuition, expertise in modular design patterns, proficient with UML diagrams and ERD data schemas.
- **Weaknesses:** Less preference for rapid UI design iterations; focuses primarily on structural soundness.
- **Agreed Contributions:** Defining core system architecture, data models, component interaction interfaces, and ensuring structural scalability across sprints.

3. Bùi Đỗ Thiên Ân (25125050) - Algorithm & Logic Developer

- **Strengths:** Exceptional mathematical foundations, deep understanding of discrete math, optimization, and complex algorithmic constraints.
- **Weaknesses:** Deep theoretical focus requiring collaborative bridge-building when integrating algorithms with live APIs.
- **Agreed Contributions:** Designing and optimizing computational algorithms, formulating mathematical models, and validating computational complexity.

4. Nguyễn Đức Huy (25125014) - Front-end Developer

- **Strengths:** Creative UI/UX styling, proficiency in responsive web frameworks, interactive dashboard design, and user journey flow.
- **Weaknesses:** Less familiarity with database indexing and server-side concurrency tuning.
- **Agreed Contributions:** Designing responsive user interfaces, implementing interactive visualizations, ensuring cross-platform usability, and integrating front-end views with backend APIs.

5. Lý Trần Thái (25125048) - Core Developer & Data Integration (Author)

- **Strengths:** Versatile full-stack implementation, strong grasp of sensor data pipelines, multimodal feature extraction, and strict Git version control hygiene.
- **Weaknesses:** Tendency toward perfectionism in data pipelines; prefers programmatic data contracts over visual styling.
- **Agreed Contributions:** Developing core logic modules, building data ingestion and preprocessing pipelines, writing automated integration tests, and maintaining documentation and Git history.

6. Trần Đăng Khoa (25125056) - Back-end Developer

- **Strengths:** Robust server-side programming, RESTful API design, database connection pooling, and endpoint security.
- **Weaknesses:** Minimal interest in frontend CSS layout styling.
- **Agreed Contributions:** Implementing backend service endpoints, managing data persistence, handling server-side validation, and optimizing API response latency.

7. Ngô Tấn Đạt (23125054) - Security & Data Engineer

- **Strengths:** Deep knowledge of information security, defensive programming, data sanitization, authentication tokens, and vulnerability prevention.
- **Weaknesses:** Prefers backend security architecture over visual design tasks.
- **Agreed Contributions:** Conducting security audits on code commits, ensuring safe storage and transmission of credentials, managing data integrity, and configuring secure deployment pipelines.

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4. Team Synergy & Collaboration Agreement

To guarantee accountability and harmonious progress:

1. **Complementary Skill Matching:** Front-end developers work in tandem with back-end engineers using pre-agreed API mock contracts. The algorithm engineer collaborates directly with the core data integration engineer to ensure algorithms operate on sanitized real-world data structures.
2. **Support Structure:** When a member encounters a technical roadblock lasting more than 24 hours, they raise a flag in the team channel for immediate peer pairing.
3. **Commitment to Quality:** Every member agrees to uphold the team contract standards, submit clean code with explanatory comments, and respect agreed internal deadlines.